Report_DEB

1. Importation des données

```
file_path <- here::here("mod/Correction_X/C_DEB_SAAWYe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```
NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 =
seeds <- c(C1 = 3330, C2 = 6666, C3 = 2212)

text_priors <- '
Distrib (Fm,           TruncNormal,          0.1,      0.2,  0.0001,      0.8);
Distrib (pAm,          TruncNormal_cv,       100,      0.2,    40,    2000);
Distrib (r_pAm_pM,     TruncNormal_cv,       0.8,      0.1,    0.1,      2);
Distrib (v,            TruncNormal_cv,   0.18505,      0.1,  0.001,    2);
Distrib (kap,          Uniform,              0,     1);
Distrib (kapA,         Uniform,              0.1,     0.9);
Distrib (Ehp,          TruncNormal_cv,        100,      0.2,    40,    300);
Distrib (E_coc,        TruncNormal_cv,         30,      0.2,     1,    300);
Distrib (rOM_ClxHorse, LogUniform,        0.00001,      0.3);
Distrib (dv,           TruncNormal_cv,          2,      0.5,     0,     10);
Distrib (wE,           LogUniform,     0.000000001,     0.1);

# Calcul des vraissemblances
Distrib (Sigma_W, TruncNormal,  5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,       Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

#f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbI

# ./mcsim.DEBcali DEB_C1.in
```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

```
                 Fm.1.    pAm.1. r_pAm_pM.1.       v.1.     kap.1.    kapA.1.
Result_mode 0.13600500 290.56400  0.25904500 0.20382100 0.92661900 0.43464100
```

```
2.5%        0.09732950 268.17300  0.18065300 0.17036400 0.87217045 0.31756500
97.5%       0.34524300 318.83800  0.32975885 0.23642085 0.93221400 0.53910100
Min         0.07270070  90.58550  0.12362600 0.14469400 0.01107090 0.15520200
Max         0.76935100 338.23800  0.87744000 0.28604400 0.94057900 0.60051000
Result_mean 0.17979241 293.10605  0.26257666 0.20343582 0.91103390 0.44155128
Result_sd   0.06622923  12.99627  0.03778418 0.01676382 0.01617038 0.05450482
                Ehp.1.  E_coc.1. rOM_ClxHorse.1.     dv.1.        wE.1.
Result_mode  40.936000 14.311900     0.006666430 7.2878900 1.141450e-09
2.5%         40.236500 10.661255     0.003597068 6.0831300 1.204910e-09
97.5%        67.701800 17.764800     0.009730131 8.7783800 8.025170e-07
Min          40.002500  9.139400     0.000125481 0.5220400 1.024840e-09
Max         107.006000 32.418600     0.013618700 9.9407700 2.155700e-06
Result_mean  47.985358 13.748988     0.006135416 7.4062428 1.083870e-07
Result_sd     7.299482  1.768379     0.001643091 0.7037093 2.184455e-07
            Sigma_W.1.
Result_mode 0.16780800
2.5%        0.15323300
97.5%       0.19621665
Min         0.13833400
Max         5.67894000
Result_mean 0.17360365
Result_sd   0.04742979

Number of observations, n =  277
Number of parameters, k =  11
Log-likelihood, LL =  -78.61879
BIC =  219.1018


Potential scale reduction factors:

                Point est. Upper C.I.
dv.1.                 1.00       1.01
E_coc.1.              1.01       1.03
Ehp.1.                1.27       1.83
Fm.1.                 1.01       1.04
kap.1.                1.15       1.44
kapA.1.               1.02       1.03
pAm.1.                1.01       1.04
r_pAm_pM.1.           1.02       1.02
rOM_ClxHorse.1.       1.01       1.02
Sigma_W.1.            1.00       1.01
```

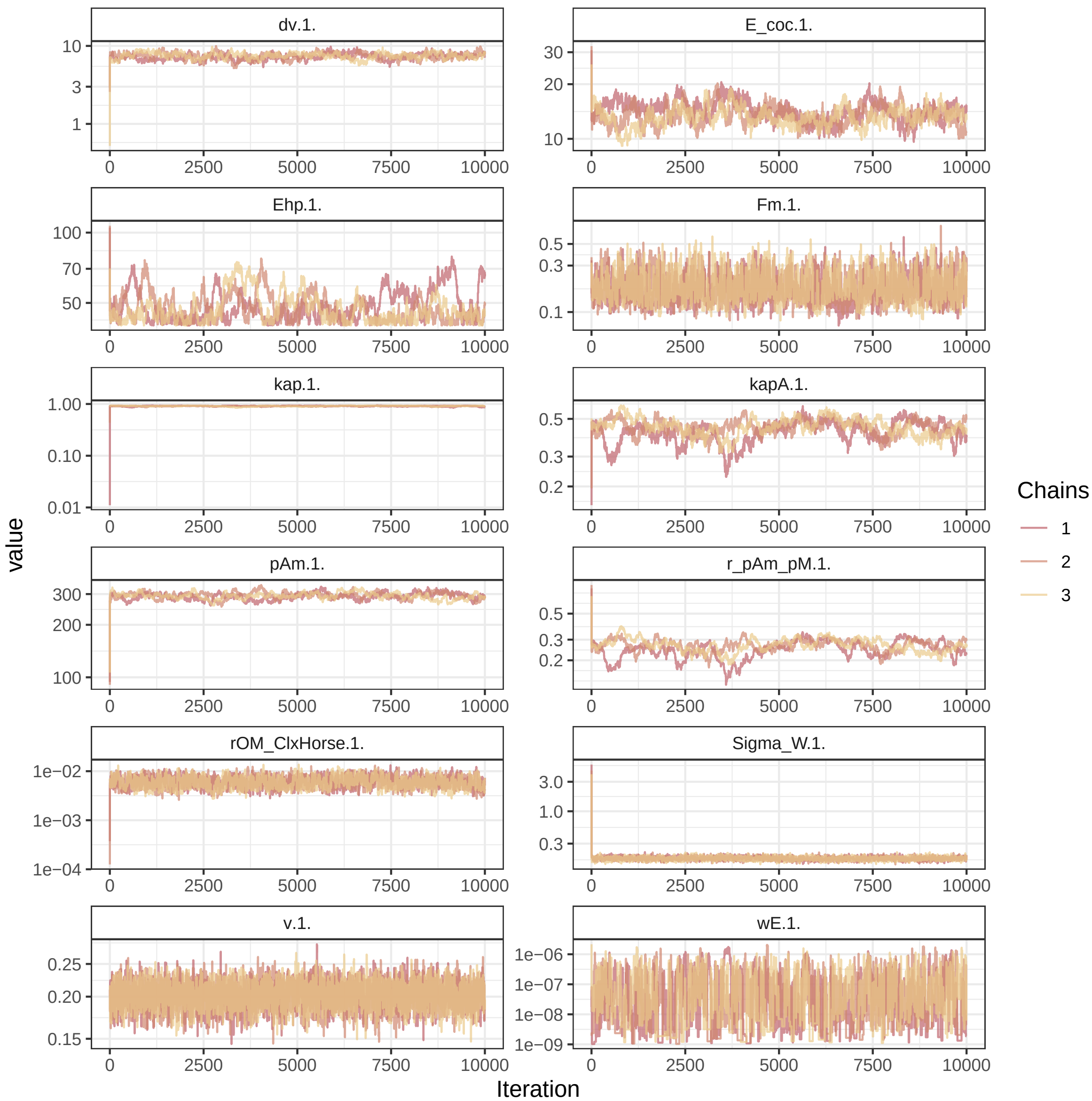


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

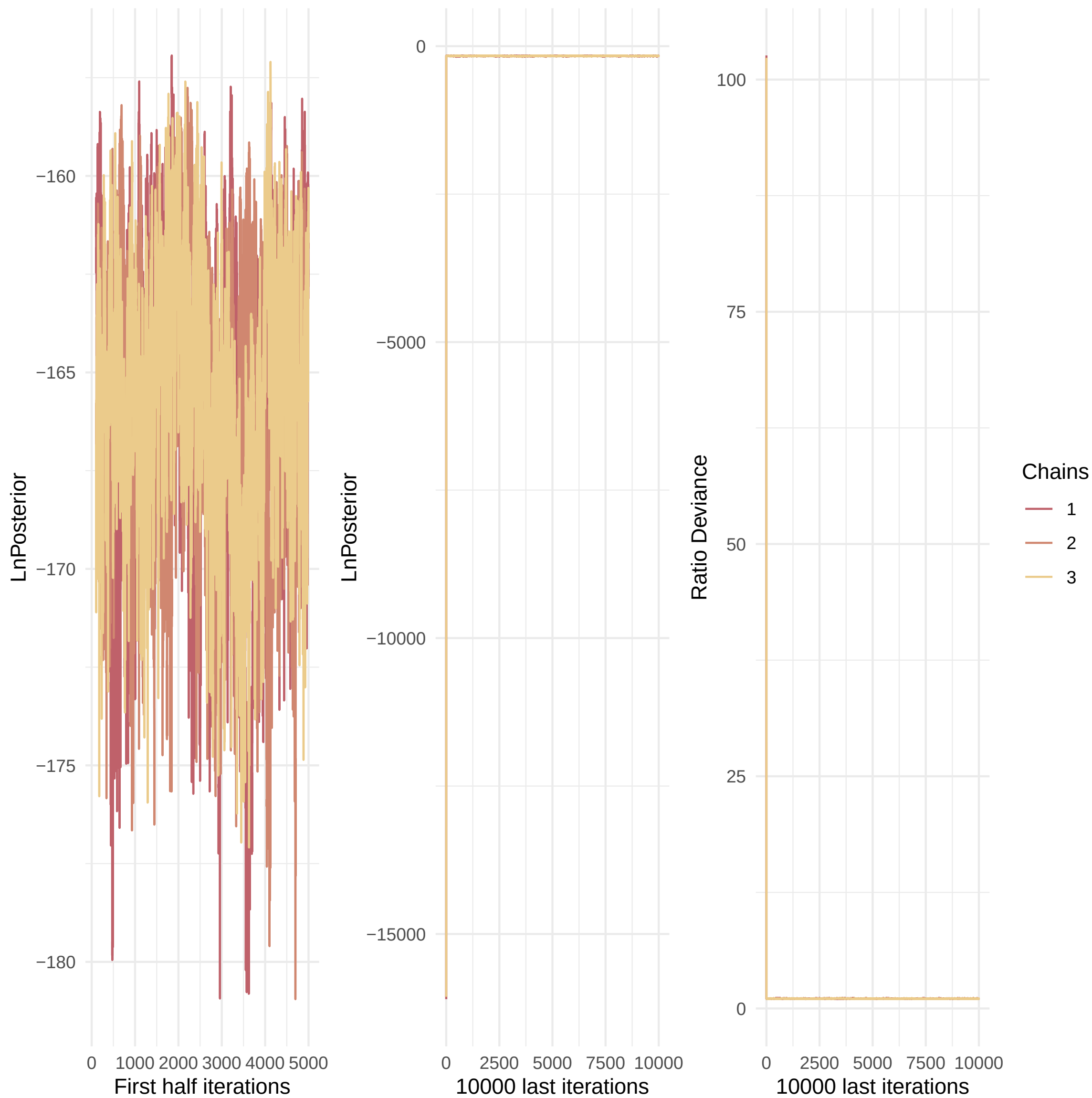


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

```
v.1.                    1.00       1.00
wE.1.                   1.01       1.02

Multivariate psrf

1.15
```

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE])  # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

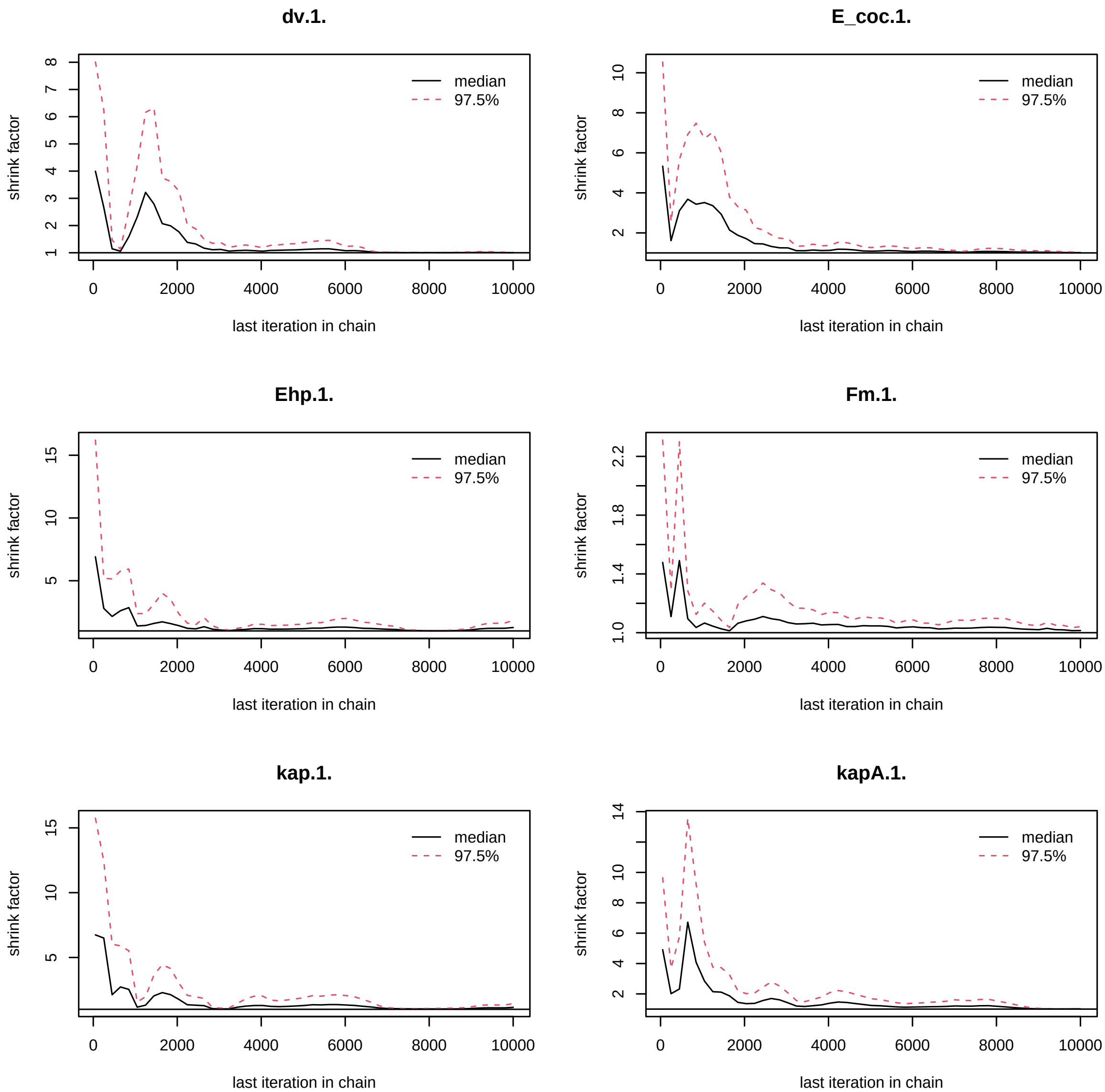


Figure 3: Chains convergence.

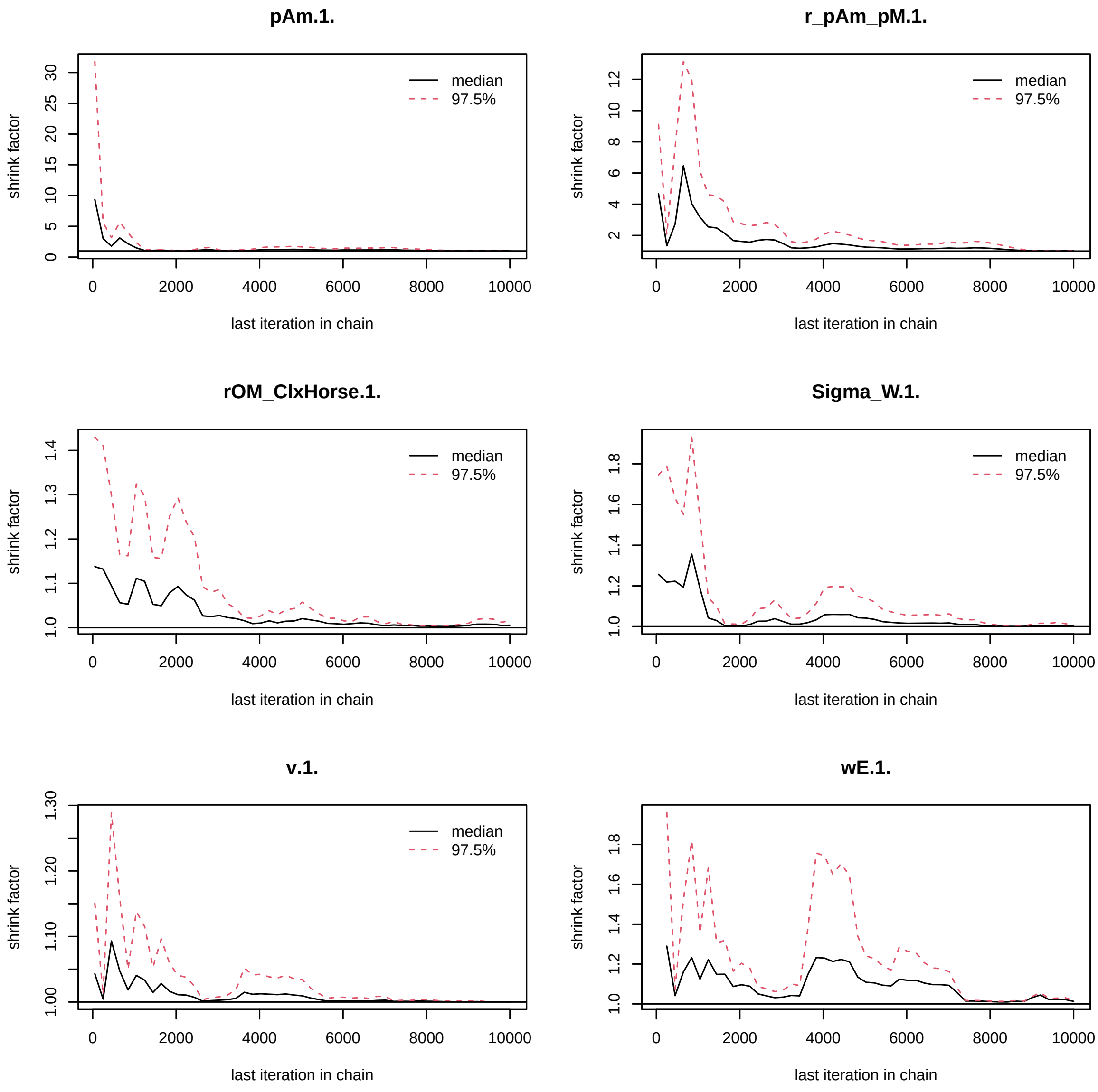


Figure 4: Chains convergence.

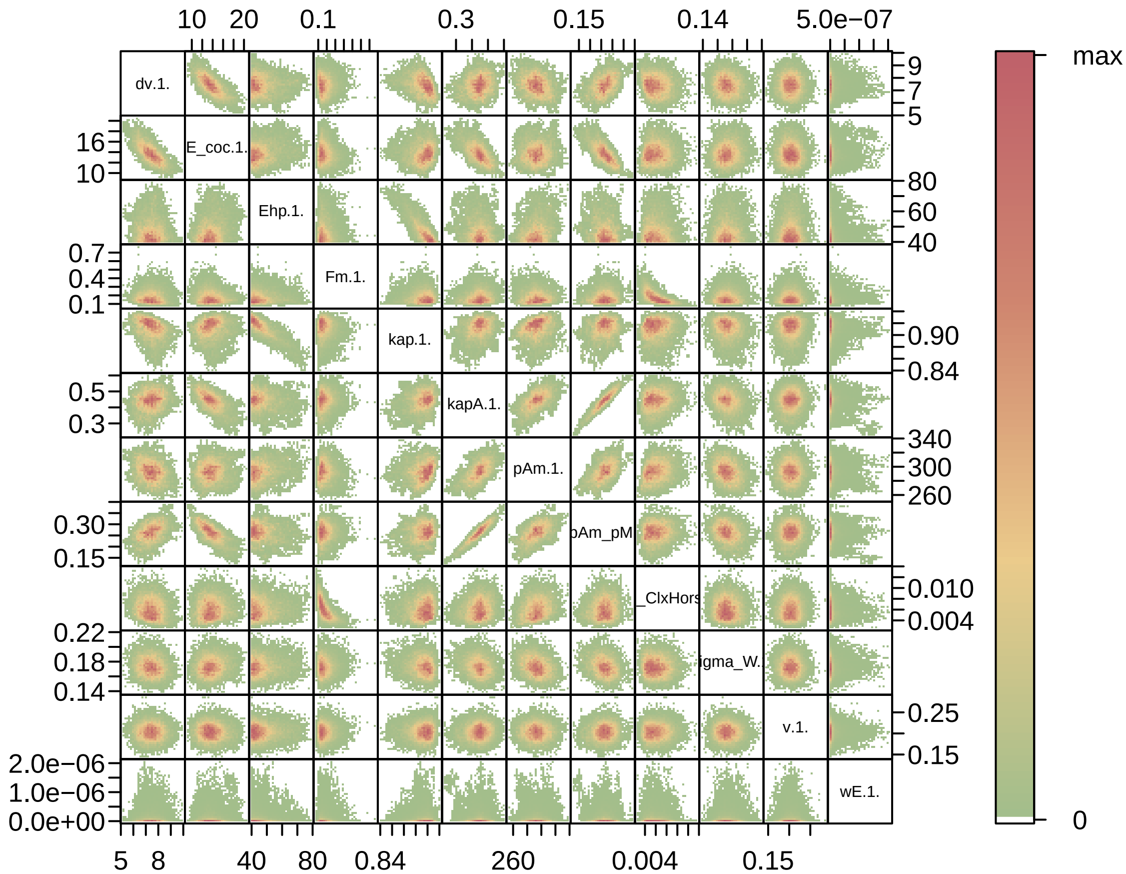


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

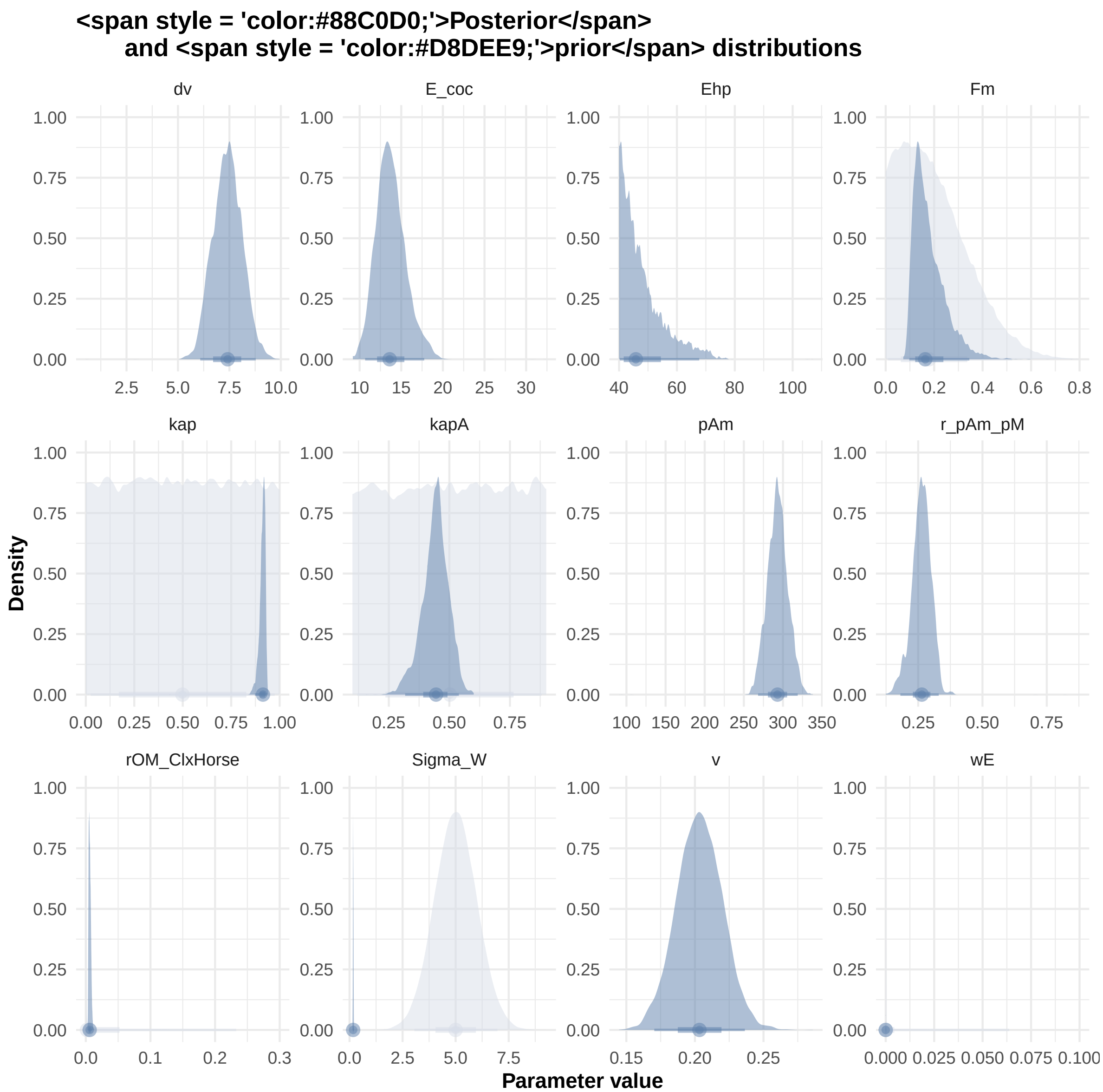


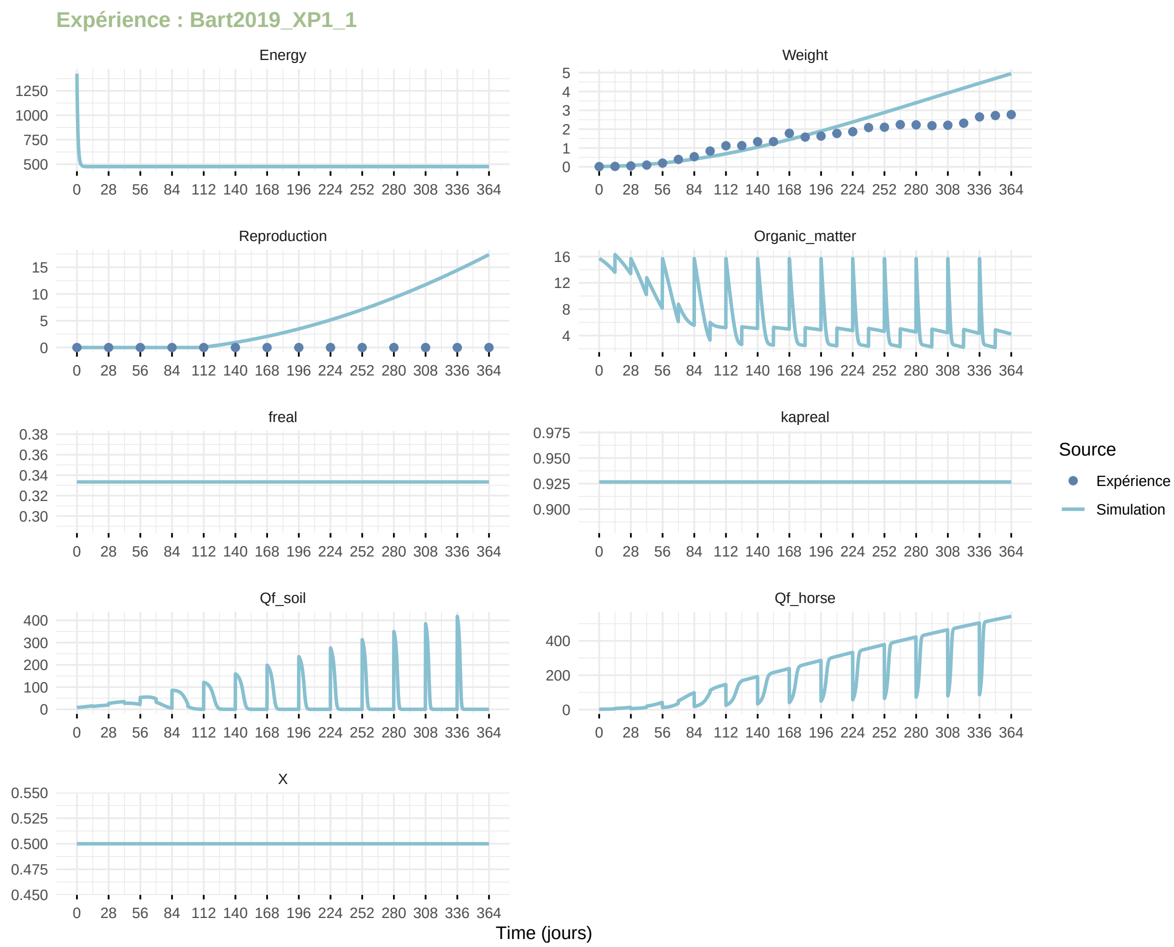
Figure 6: Distributions of the priors and the posteriors.

```
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\.1\\.$|\\.1\\.", "", param),
    line  = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
    Length_struct,
    Reproduction,
    Energy,
    Organic_matter,
    kapreal,
    freal,
    Qf_soil,
    Qf_horse,
    X"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_prin
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pri

# ./mcsim.DEBcali DEB_sim_full.in
```

```r
##############################
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#############################

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
```

```
Warning: `read_table2()` was deprecated in readr 2.0.0.
i Please use `read_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpe
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
```

"many-to-many"` to silence this warning.



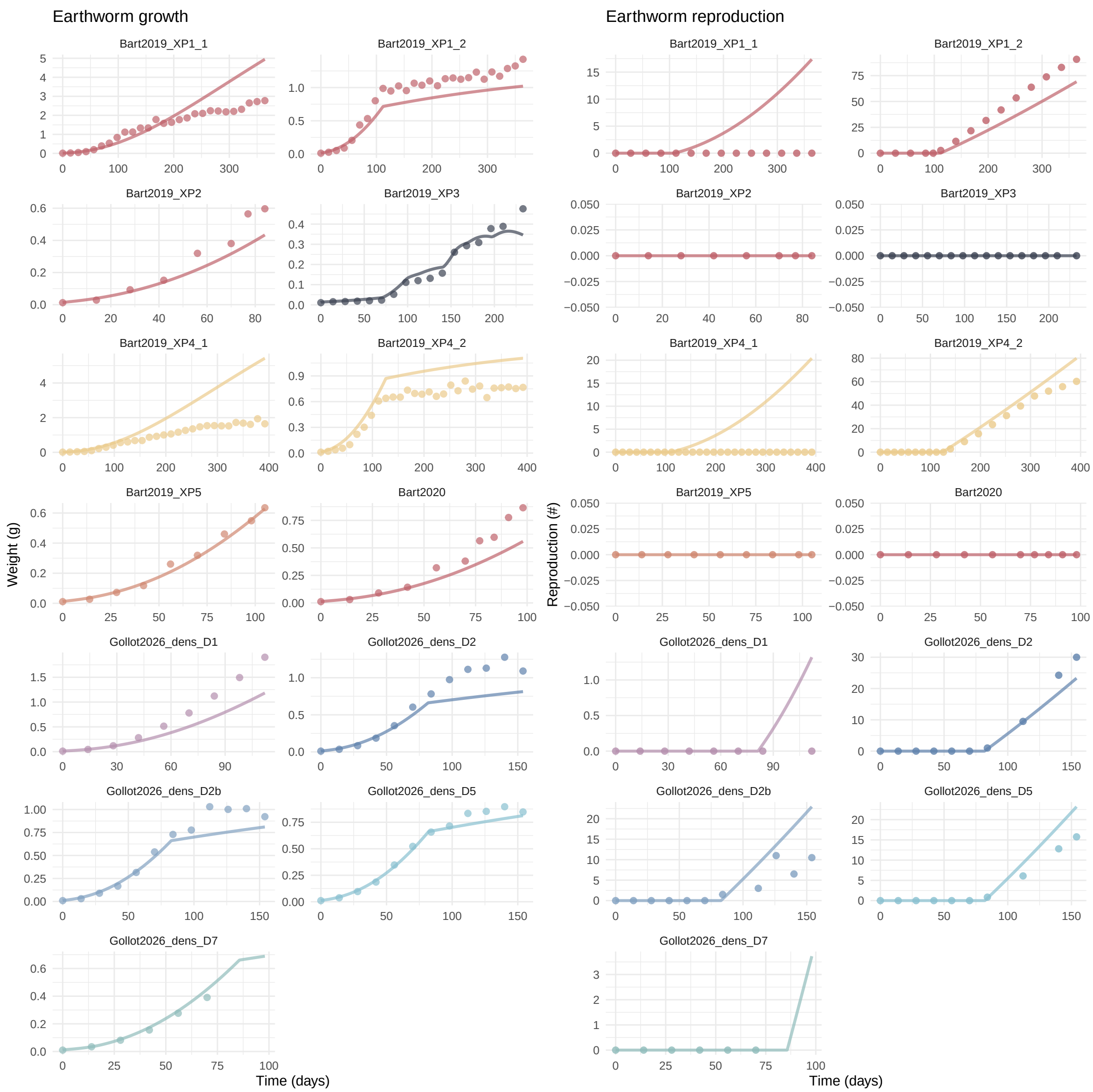
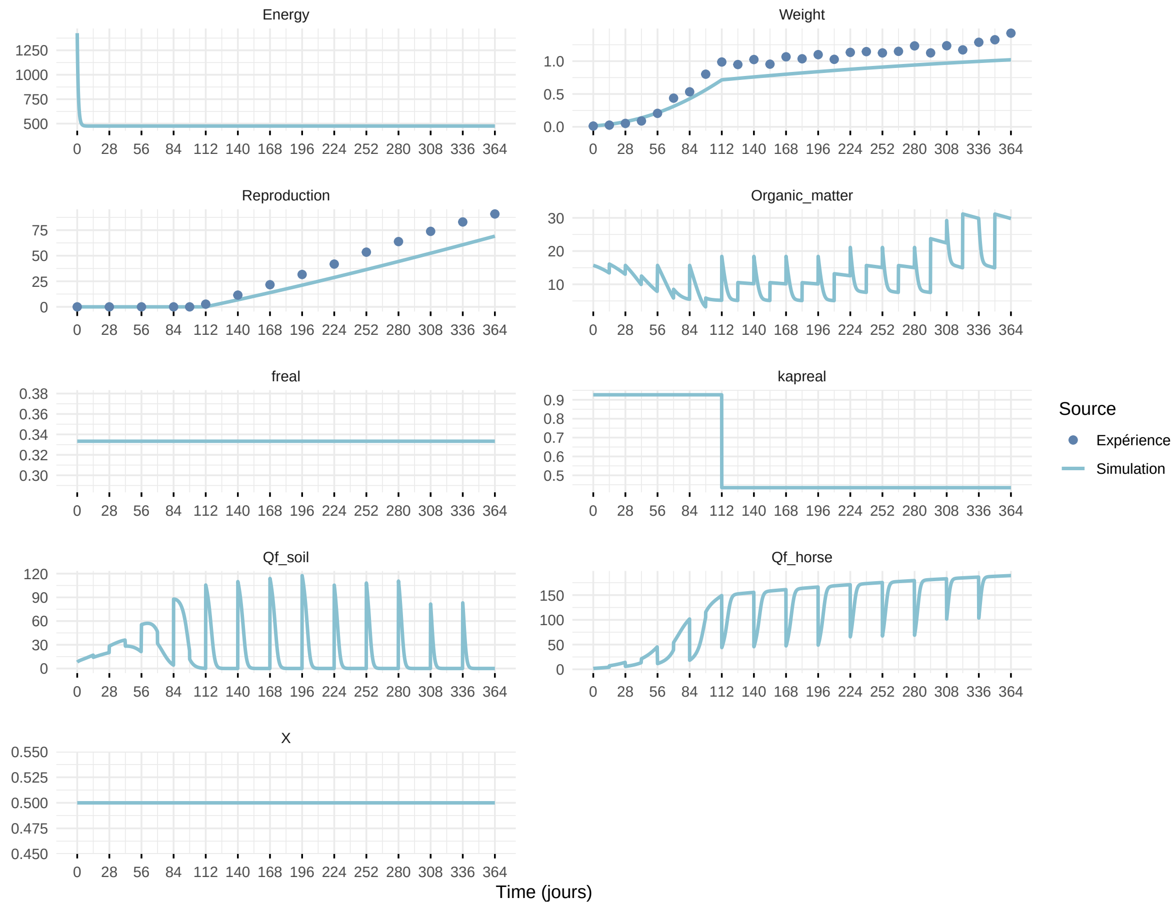
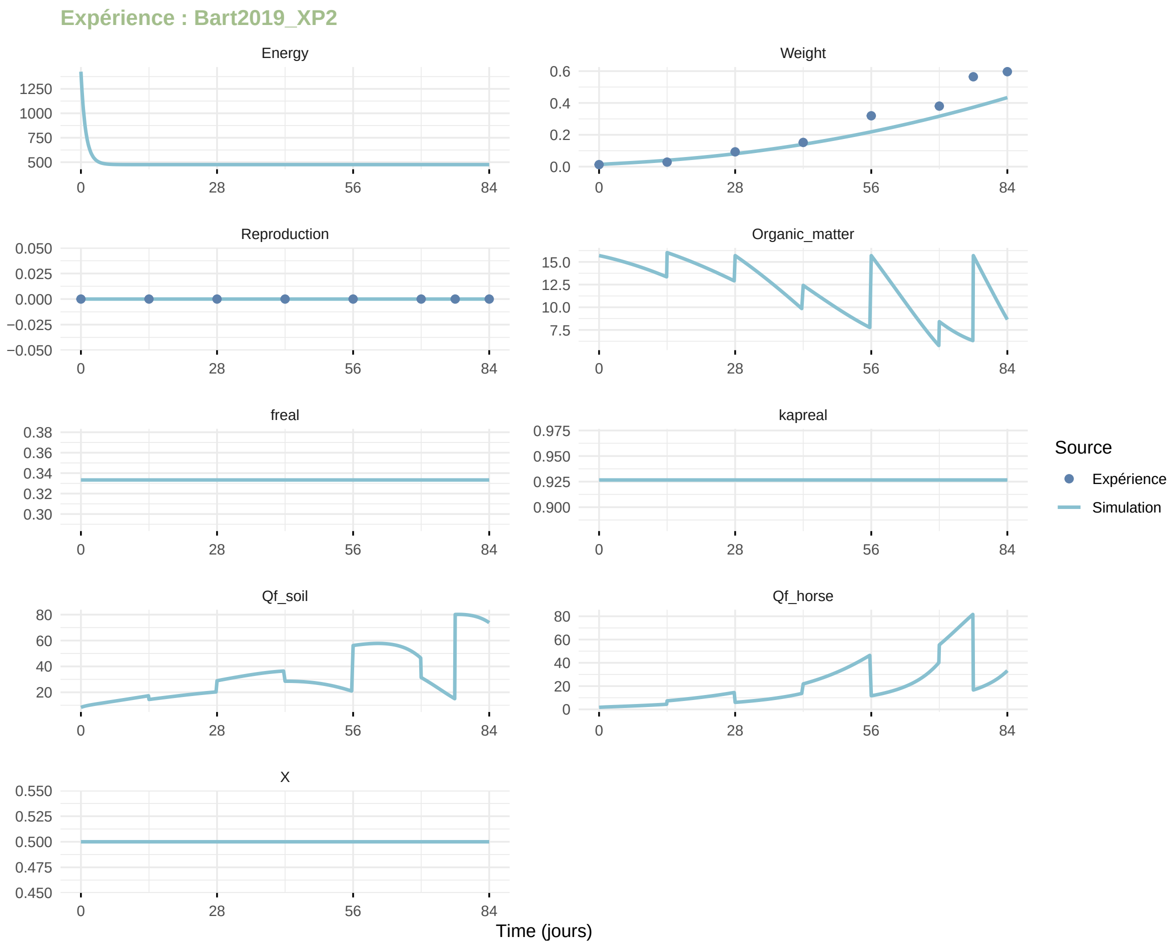


Figure 7: Simulations Weight and Reproduction.

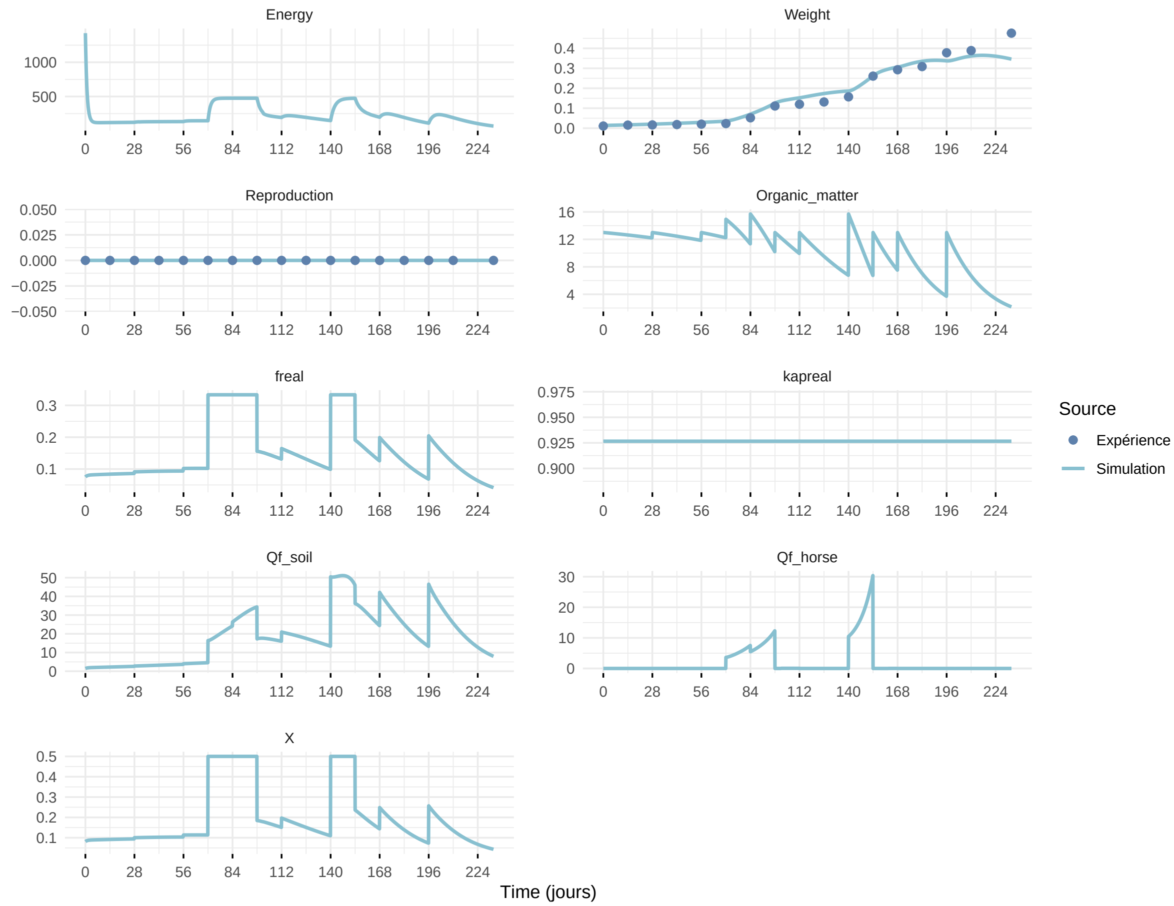
Expérience : Bart2019_XP1_2

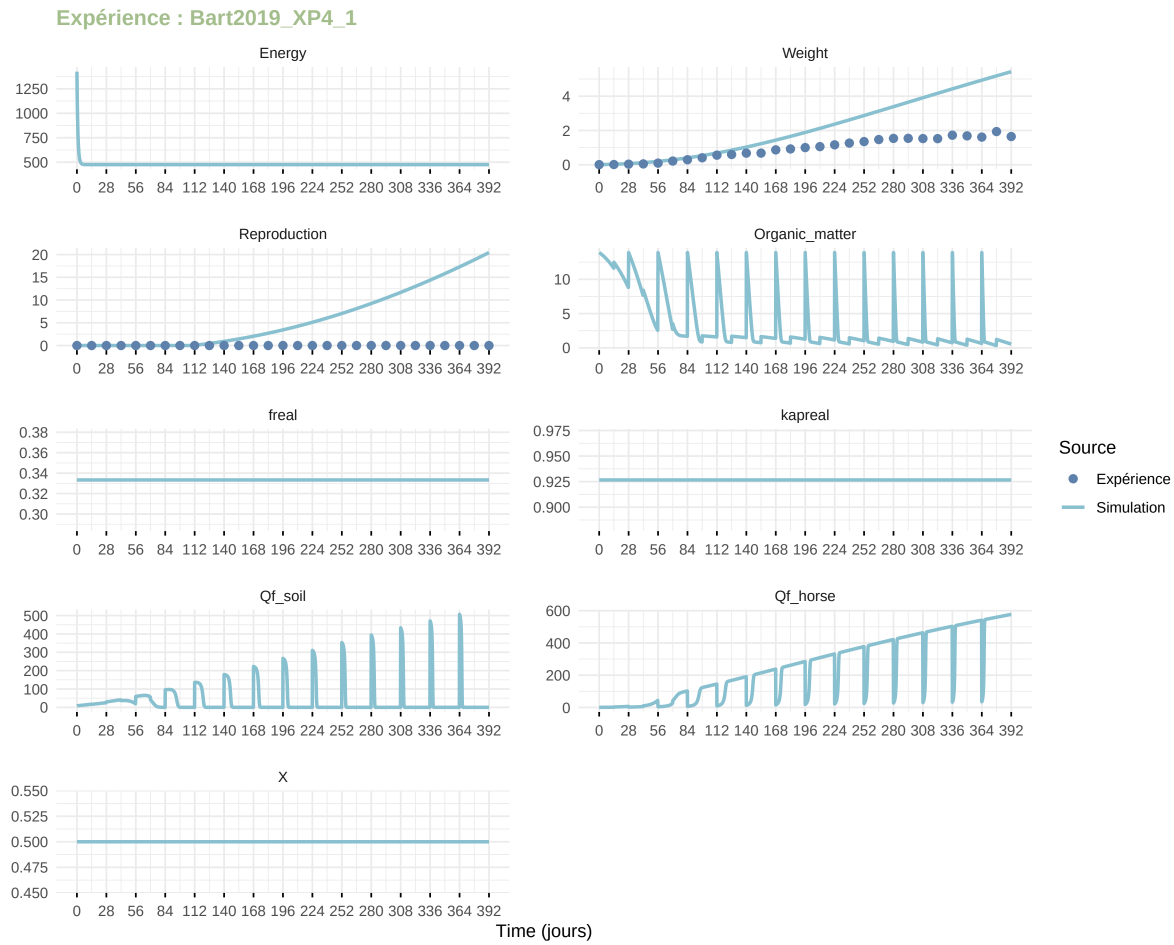




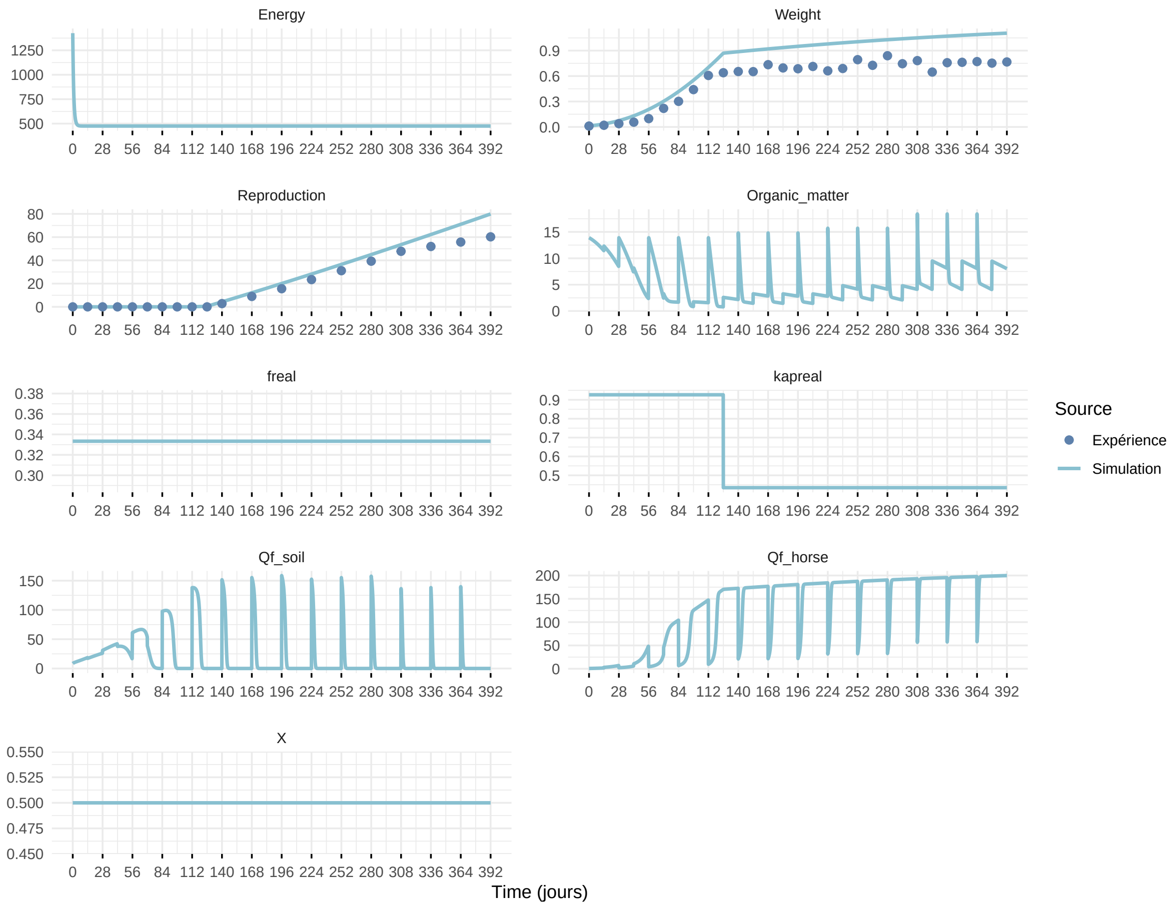
Expérience : Bart2019_XP2
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Source
Expérience
Simulation
Time (jours)

Expérience : Bart2019_XP3

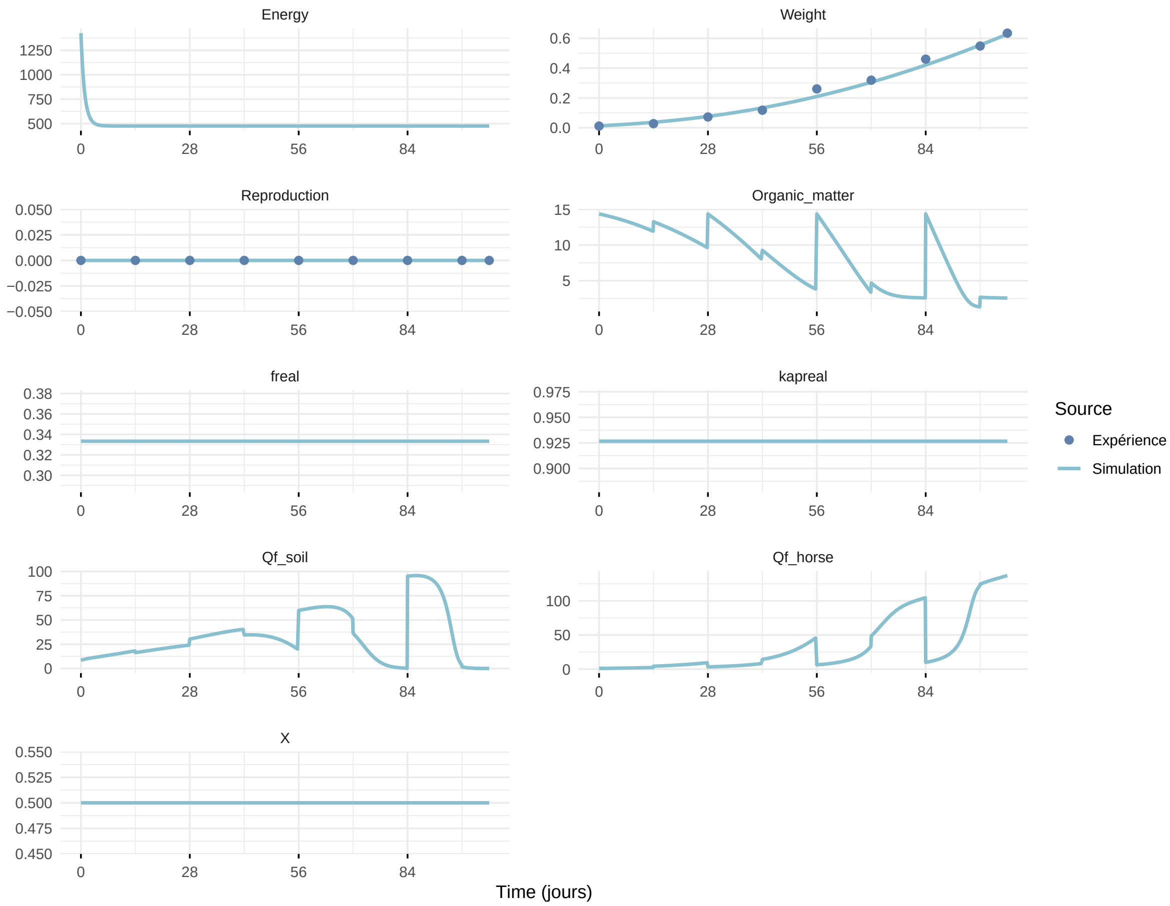


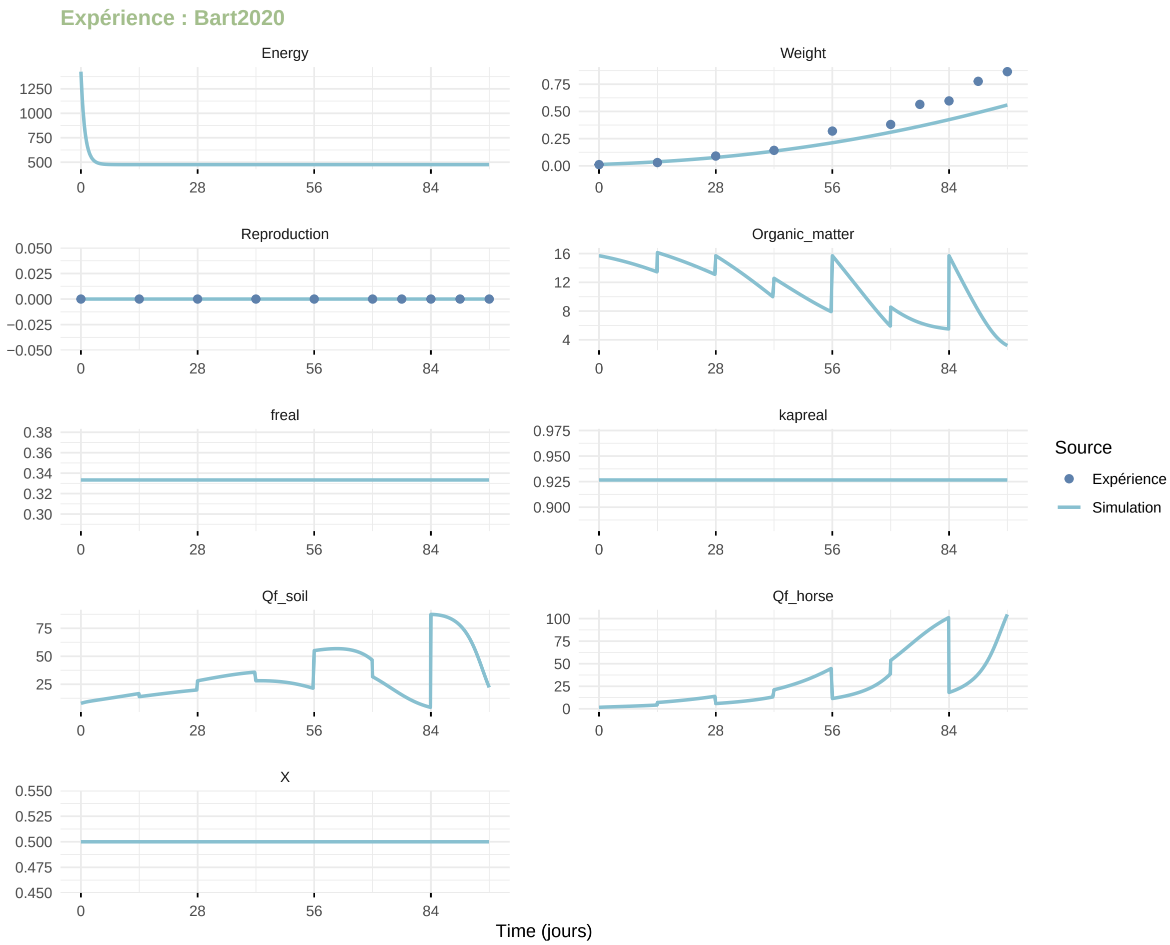

Expérience : Bart2019_XP4_1
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
0 28 56 84 112 140 168 196 224 252 280 308 336 364 392
Time (jours)
Source
Expérience
Simulation

Expérience : Bart2019_XP4_2

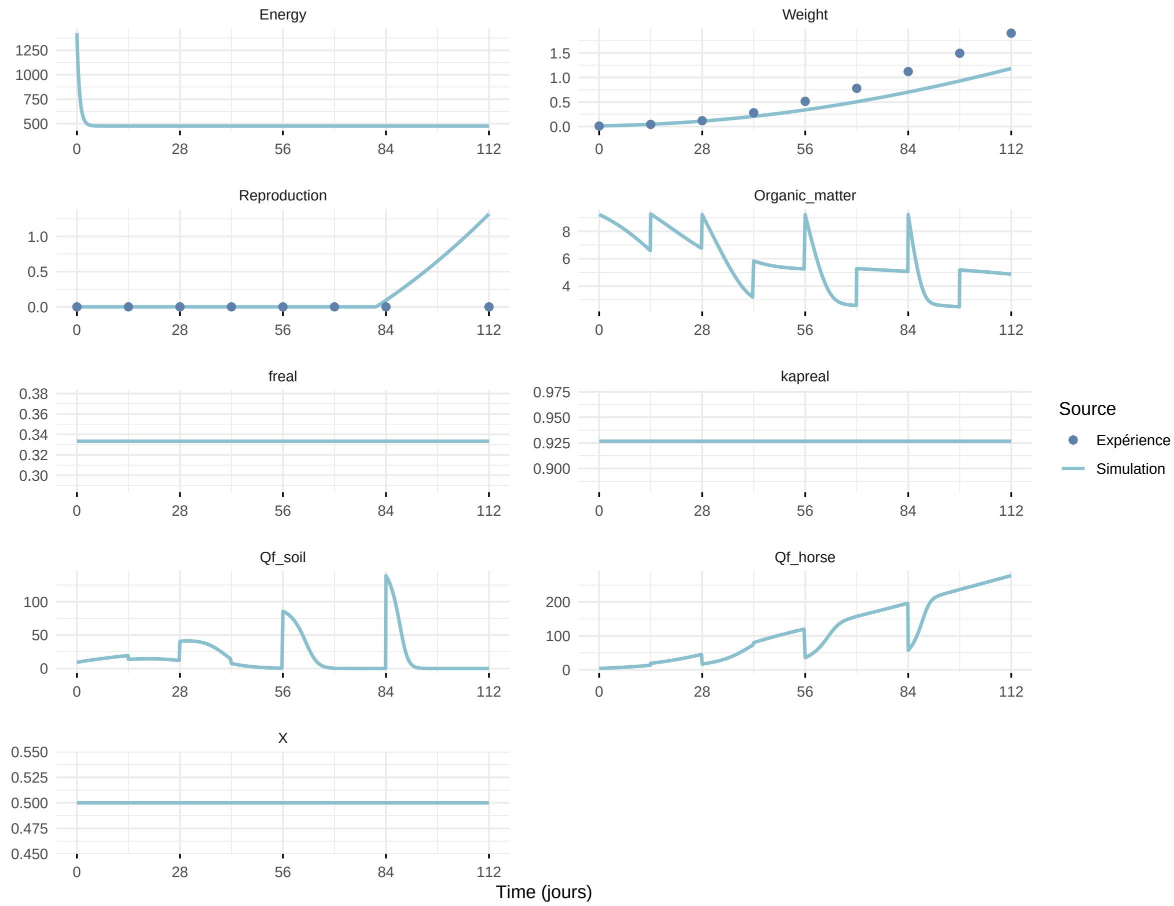


Expérience : Bart2019_XP5

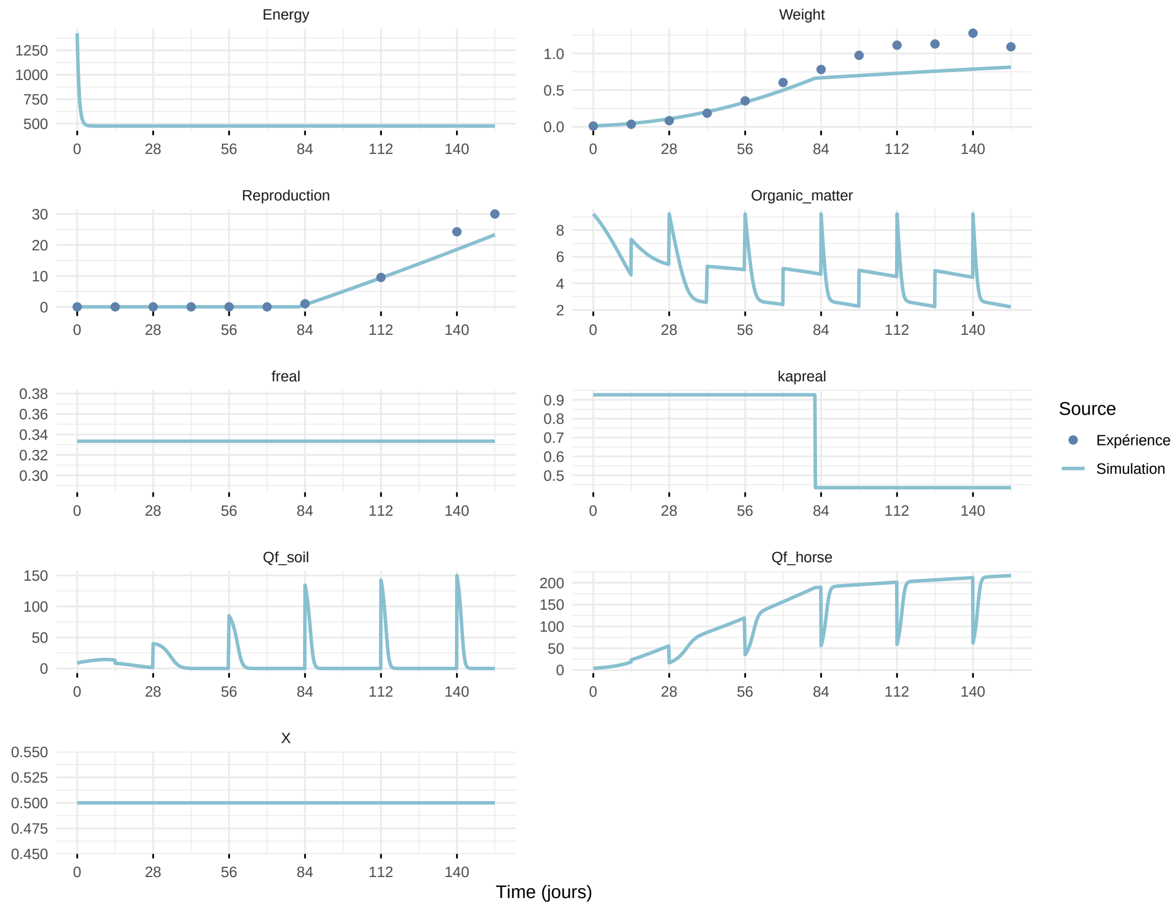



Expérience : Bart2020
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Source
Expérience
Simulation
Time (jours)

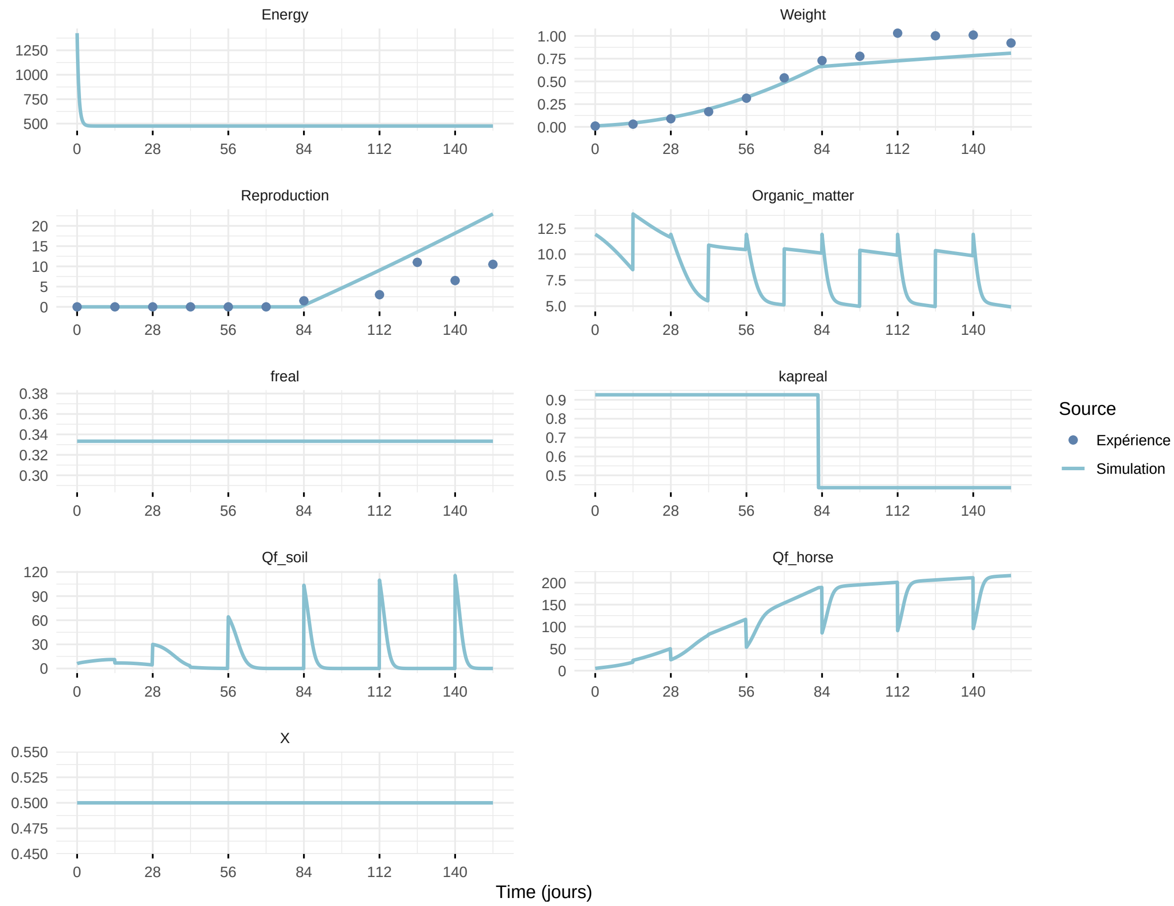
Expérience : Gollot2026_dens_D1



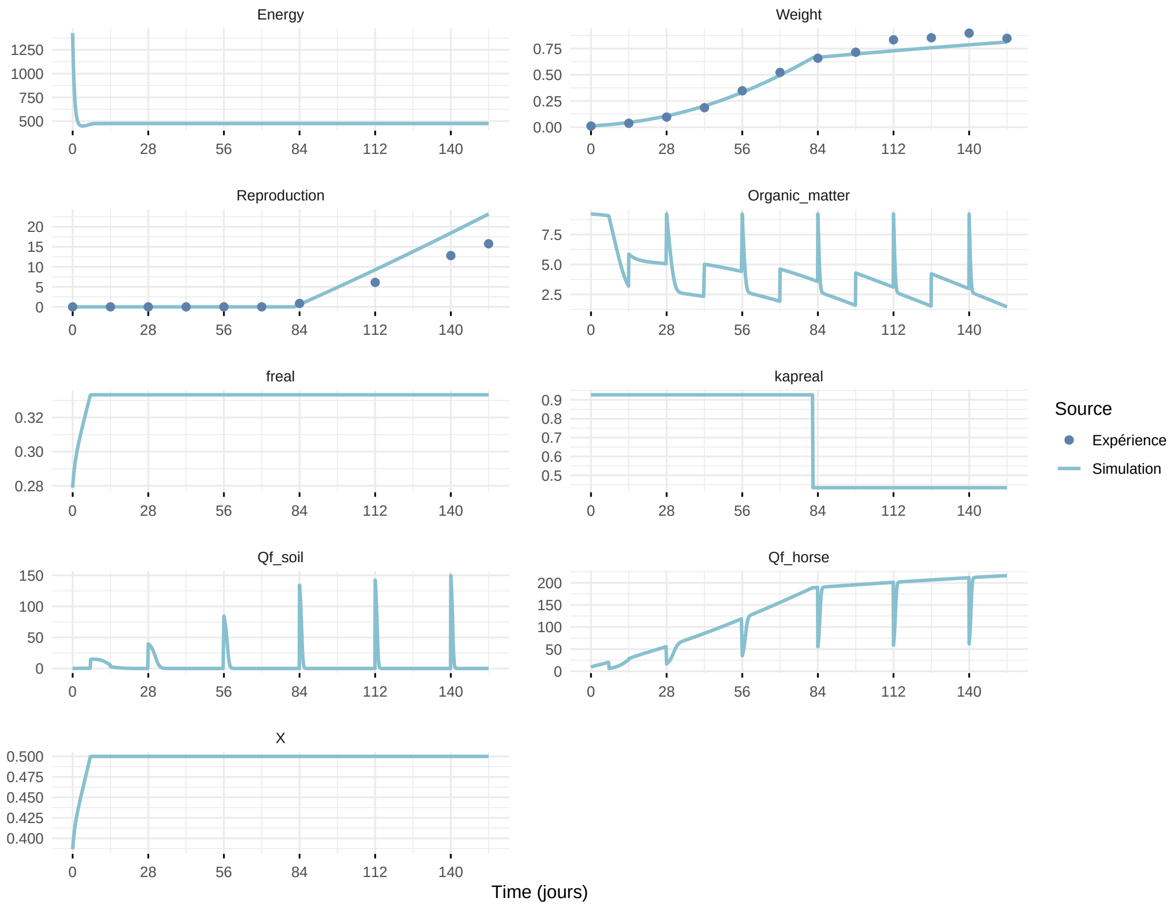
Expérience : Gollot2026_dens_D2

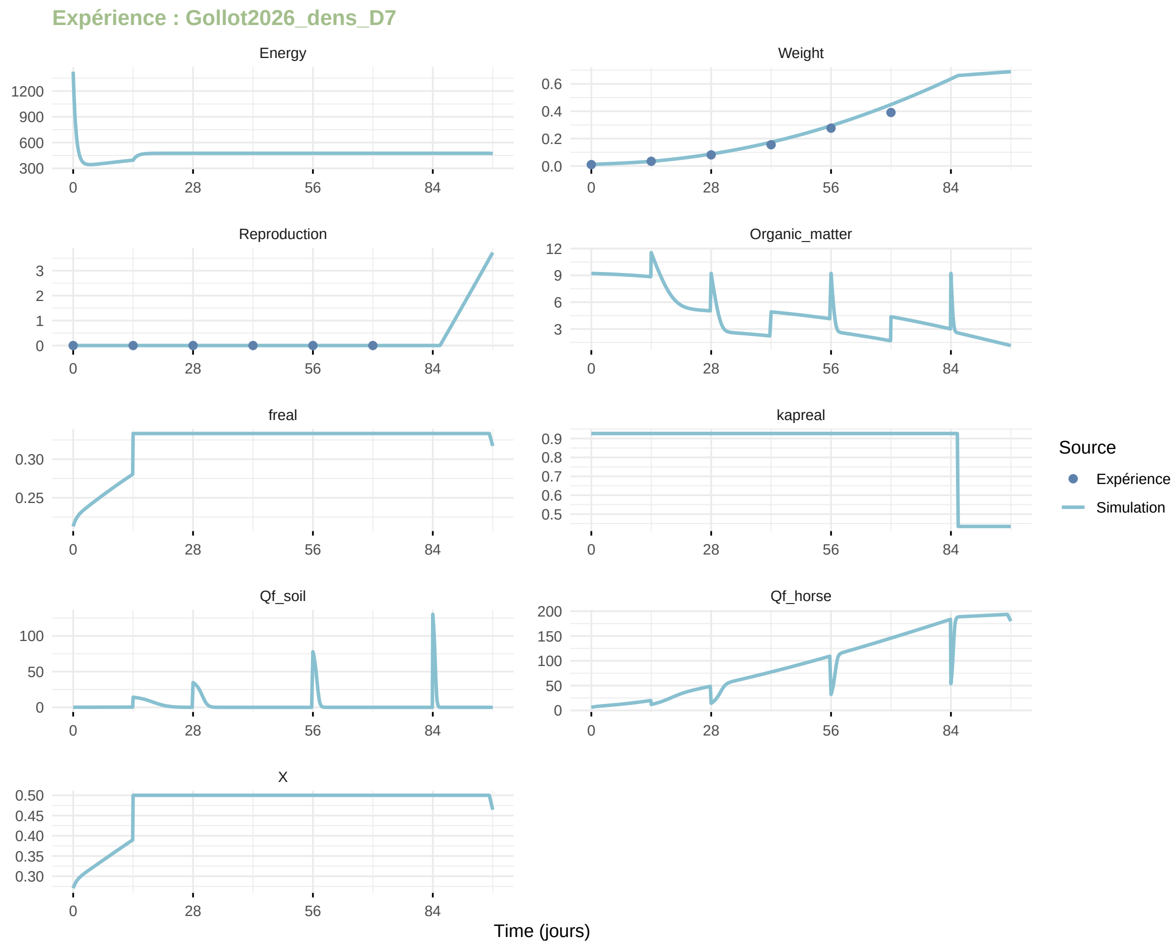


Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5




Expérience : Gollot2026_dens_D7
Energy
Weight
Reproduction
Organic_matter
freal
kapreal
Qf_soil
Qf_horse
X
Source
Expérience
Simulation
Time (jours)